

Statistical analysis plan (SAP)

Write the analysis down before you see the data

Foundations of Clinical Research · Sessions 5–6 (6.4) · Template version 1.0

How to use this template

- Replace every [grey italic text in brackets] with your own text. Use Ctrl + H → Find “[” to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded “Example” boxes are there to help you. Delete them as you write: Home → Styles → right-click “Guidance” (or “Example”) → Select All → Delete.
- Finish and sign the SAP before the database is locked (and, in a trial, before unblinding). Analyses that were not planned can still be done, but must be reported as exploratory.
- The examples use the running example as a two-arm trial: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?* Notes marked “Observational study” show what changes for a cohort.
- Write it with your statistician. In this course you need to understand each choice; the instructor runs the analyses.
- **Worked example:** a complete SAP for the course dataset (FLUID-ICU, positive fluid balance by day 3 and 28-day mortality) is in Course/Data/SAP_FLUID-ICU.md. Compare your plan with it.

Study title: [Full title]
Protocol version: [Version and date]
SAP version / date: [1.0] · [DD/MM/YYYY]
Written by: [Name(s) and role(s)]

1 Study summary and objectives

Copy from the protocol: design, population, intervention/exposure and comparator, and the primary and secondary objectives. Every analysis below must link to one objective.

Example: Pilot randomised trial, 1:1, adults with septic shock. Primary objective: compare 28-day all-cause mortality between a restrictive fluid strategy and usual care. Secondary: cumulative fluid balance at day 5; renal replacement therapy; time to death up to day 28.

[Summary and objectives]

2 Outcomes and their definitions

Define each outcome exactly as it will be computed from the database: variable name, type, unit and time point. One primary outcome.

Outcome	Type	Definition (variables)	Time point
Primary: all-cause mortality	Binary	vital_d28 = 1 (dead)	Day 28
Cumulative fluid balance	Continuous (ml/kg)	cfb_kg_d5 = $\Sigma(\text{fin} - \text{fout})$ days 1–5 ÷ weight	Day 5
Time to death	Time-to-event	death_dt – enrol_dt; censored at day 28	Up to day 28
[Outcome]	[Type]	[Definition]	[Time point]
[Outcome]	[Type]	[Definition]	[Time point]

3 Analysis populations

Who is included in each analysis. Trials: the intention-to-treat (ITT) population is everyone randomised, analysed in the group they were allocated to. Observational studies: everyone meeting the eligibility criteria, with reasons for any exclusion in the flow diagram.

Population	Definition	Used for
Intention-to-treat (ITT)	[All randomised participants, as allocated]	[Primary analysis]
Per-protocol	[Participants who received the allocated strategy as defined in the protocol]	[Sensitivity analysis]
Safety	[All participants who received any study intervention, as treated]	[Adverse events]

4 Descriptive analysis (Table 1)

Describe baseline characteristics by group (see the Table 1 shell in section 13). Continuous variables: mean (SD) if roughly symmetric, otherwise median (IQR). Categorical: n (%). In a trial, do NOT test baseline differences with p-values — any difference is due to chance. Report the number with missing data for each variable.

Example: Age, sex, weight, source of infection, comorbidities, SOFA score and lactate summarised per arm; lactate is skewed, so median (IQR).

[Our descriptive plan]

5 Primary analysis

One analysis, fully specified: the test or model, the effect measure with its 95% confidence interval, any adjustment, and the significance level. The p-value supports the effect estimate; it does not replace it.

Outcome: [Primary outcome and population]
Method: [Test or model]
Effect measure: [e.g. risk ratio, risk difference, odds ratio, hazard ratio, mean difference]
Adjustment: [Variables, and why]
Significance: [Two-sided, 5% level]

Example: 28-day mortality in the ITT population compared with a chi-square test. Effect reported as risk ratio and risk difference with 95% CI. Supportive analysis: logistic regression adjusted for the stratification variable (baseline lactate > 4 mmol/L), reported as odds ratio (95% CI).

6 Secondary analyses

For each secondary outcome: summary, test or model and effect measure. Say clearly that secondary results are supportive or exploratory.

Example: Fluid balance: median (IQR) per arm, Mann–Whitney U test, difference in medians with 95% CI. Time to death: Kaplan–Meier curves, log-rank test, Cox model hazard ratio (95% CI), proportional hazards checked. Renal replacement therapy: risk ratio (95% CI).

[Our secondary analyses]

7 Confounders and adjustment

Trials: adjust only for stratification variables and a few pre-specified strong prognostic factors. *Observational study:* list the confounders chosen IN ADVANCE from subject knowledge (a simple causal diagram helps), not by stepwise selection. Keep about 10 outcome events per variable in the model.

Example: Observational version of the running example: sicker patients receive more fluid AND die more (confounding by indication), so the model adjusts for age, SOFA score, lactate and source of infection — with about 68 expected deaths, at most 6–7 variables.

[Our adjustment plan]

8 Subgroups and interaction (pre-specified)

List only a few subgroups, with the expected direction and a reason. Test whether the effect differs with an interaction test — not by comparing p-values in each subgroup. Present a forest plot and call the results exploratory.

Example: Baseline lactate (≤ 4 vs > 4 mmol/L) and SOFA score (≤ 10 vs > 10): treatment $\times$ subgroup interaction term in the logistic model.

[Our subgroups]

9 Multiple comparisons

Many tests mean some “significant” results by chance (the jelly-bean problem). Protect yourself with ONE primary outcome. Say how secondary outcomes are handled: interpreted as exploratory without adjustment, or adjusted (e.g. Holm or Bonferroni).

Example: One primary outcome, tested at 5%. Secondary outcomes reported with 95% CIs and interpreted as exploratory; no adjustment for multiplicity.

[Our approach]

10 Missing data

Plan four things. (1) How missingness will be assessed and reported: n (%) missing per variable and by group. (2) The likely mechanism: missing completely at random (MCAR), at random given observed data (MAR), or not at random (MNAR). (3) The main approach: complete-case analysis, or multiple imputation if more than about 5% is missing and MAR is plausible. (4) A sensitivity analysis that tests the assumption.

Assessment: [How missingness is described and explored]

Main approach: [Complete-case / multiple imputation (variables in the model, number of imputations)]

Sensitivity: [e.g. best-case / worst-case, or a different imputation model]

Example: Vital status at day 28 is expected to be missing for < 2% (phone follow-up). Main analysis: complete case. Sensitivity: missing patients assumed dead (worst case) and alive (best case). Baseline lactate, if > 5% missing, multiply imputed (20 imputations) from age, SOFA and arm.

11 Sample size

Copy the calculation from the protocol and state every assumption with its source: expected value in the comparison group, smallest important difference, power, significance level, allowance for loss.

Example: 40% vs 30% mortality, 80% power, two-sided 5%: 356 per group, about 790 in total with 10% loss. The pilot (60 patients) is not powered for mortality; it estimates recruitment and adherence.

[Our sample size]

12 Software

Name the software and version, and where the analysis code will be kept.

[e.g. R version 4.x (packages ...) / Stata / SPSS; code stored in [location] with version control]

13 Table and figure shells

Draw the empty tables and figures now. They show exactly what will be calculated, and they become the tables of the paper. Use *n* (%) and median (IQR) with units, *n* per column and footnotes.

Table 1. Baseline characteristics by group

Characteristic	Restrictive (n = __)	Usual care (n = __)
Age, years, mean (SD)		
Female sex, n (%)		
SOFA score, median (IQR)		
Lactate, mmol/L, median (IQR)		
Source of infection: lung, n (%)		
[Characteristic, unit, summary]		

Footnote: SD, standard deviation; IQR, interquartile range. Missing data: [n] for lactate.

Table 2. Primary and secondary outcomes

Outcome	Restrictive	Usual care	Effect (95% CI)	p
28-day mortality, n (%)			RR __ (__ to __)	
Cumulative fluid balance day 5, ml/kg, median (IQR)			Difference __ (__ to __)	
Renal replacement therapy, n (%)			RR __ (__ to __)	

Figures

- Figure 1 — Participant flow diagram (CONSORT for trials, STROBE for observational studies).
- Figure 2 — Kaplan–Meier curves of survival to day 28 by group, with numbers at risk.
- Figure 3 — Forest plot of pre-specified subgroups (if any).

14 Version history and sign-off before data lock

Sign before the database is locked (and before unblinding in a trial). Any later change gets a new version number, the date and the reason — and is reported in the paper.

Version	Date	Changes and reason
[1.0]	[DD/MM/YYYY]	First version
[1.1]	[DD/MM/YYYY]	[Change and reason]

Role	Name	Signature	Date (DD/MM/YYYY)
Statistician			
Principal investigator			
Other (e.g. trial steering committee)			

Database lock date: [DD/MM/YYYY]